

A Concise Tutorial on Digital and Hybrid Beamforming in Fluid Antenna Systems

1 System model

We consider downlink multiuser MISO transmission with a base station serving K single-antenna users through a fluid antenna system (FAS). The FAS aperture contains M candidate ports, and N of them are active during one transmission interval.

Main variables

- K : number of users.
- N_s : number of data streams, usually $N_s = K$.
- M : total number of candidate FAS ports.
- N : number of active FAS ports.
- N_{RF} : number of RF chains.
- $\mathbf{s} \in \mathbb{C}^{N_s \times 1}$: data stream vector, with $\mathbb{E}[\mathbf{s}\mathbf{s}^H] = \mathbf{I}$.
- $\mathcal{A} \subseteq \{1, \dots, M\}$: active-port set, with $|\mathcal{A}| = N$.
- $\mathbf{h}_k(\mathcal{A}) \in \mathbb{C}^{N \times 1}$: channel from the active FAS ports to user k .
- $\mathbf{H}(\mathcal{A}) = [\mathbf{h}_1(\mathcal{A}), \dots, \mathbf{h}_K(\mathcal{A})]^H \in \mathbb{C}^{K \times N}$: aggregate downlink channel.

This tutorial uses the **discrete-port FAS model**, because it is the cleanest model for beamforming design.

2 Classical structures

3 Digital beamforming in FAS

In digital beamforming, each active FAS port has its own RF chain, so

$$N_{\text{RF}} = N.$$

The transmitted signal is

$$\mathbf{x} = \mathbf{W}_{\text{BB}} \mathbf{s}, \quad \mathbf{W}_{\text{BB}} \in \mathbb{C}^{N \times N_s}.$$

The transmit power constraint is

$$\text{tr}(\mathbf{W}_{\text{BB}} \mathbf{W}_{\text{BB}}^H) \leq P.$$

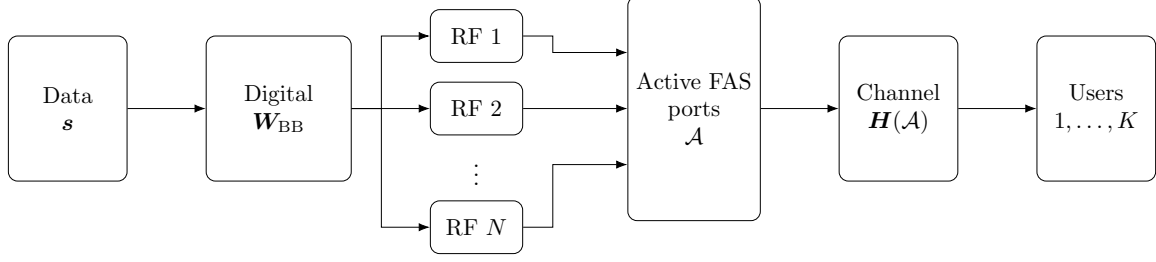


Figure 1: Digital beamforming in FAS: one RF chain per active FAS port, so $N_{\text{RF}} = N$.

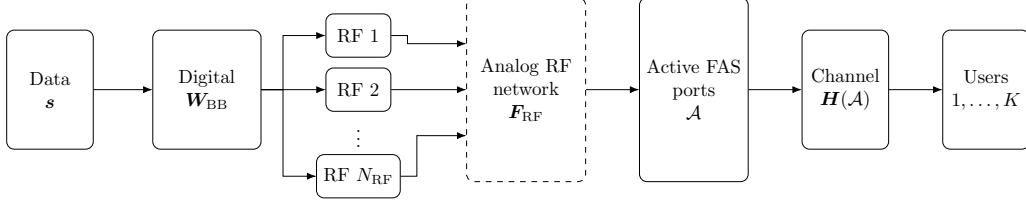


Figure 2: Hybrid beamforming in FAS: a digital stage followed by an analog RF network, with $N_s \leq N_{\text{RF}} < N$.

For user k ,

$$y_k = \mathbf{h}_k^H(\mathcal{A})\mathbf{x} + n_k = \mathbf{h}_k^H(\mathcal{A}) \sum_{i=1}^{N_s} \mathbf{w}_i s_i + n_k,$$

where $\mathbf{w}_i$ is the i th column of $\mathbf{W}_{\text{BB}}$ and $n_k \sim \mathcal{CN}(0, \sigma_k^2)$. The corresponding SINR is

$$\gamma_k = \frac{|\mathbf{h}_k^H(\mathcal{A})\mathbf{w}_k|^2}{\sum_{i \neq k} |\mathbf{h}_k^H(\mathcal{A})\mathbf{w}_i|^2 + \sigma_k^2}.$$

A standard objective is the sum-rate

$$R(\mathcal{A}, \mathbf{W}_{\text{BB}}) = \sum_{k=1}^K \log_2(1 + \gamma_k).$$

Optimization problem

$$\begin{aligned} \max_{\mathcal{A}, \mathbf{W}_{\text{BB}}} \quad & R(\mathcal{A}, \mathbf{W}_{\text{BB}}) \\ \text{s.t.} \quad & \text{tr}(\mathbf{W}_{\text{BB}} \mathbf{W}_{\text{BB}}^H) \leq P, \\ & |\mathcal{A}| = N, \quad \mathcal{A} \subseteq \{1, \dots, M\}. \end{aligned} \tag{1}$$

How to solve it

The standard method is **alternating optimization**.

1) **Fix $\mathcal{A}$, solve $\mathbf{W}_{\text{BB}}$** . This becomes a classical multiuser MISO problem. Common choices are:

- ZF,
- RZF,

- WMMSE.

For example,

$$\mathbf{W}_{\text{RZF}} = \beta \mathbf{H}^{\text{H}}(\mathcal{A}) (\mathbf{H}(\mathcal{A}) \mathbf{H}^{\text{H}}(\mathcal{A}) + \alpha \mathbf{I})^{-1},$$

where $\alpha > 0$ is the regularization parameter and β rescales the power.

2) **Fix $\mathbf{W}_{\text{BB}}$, solve $\mathcal{A}$.** This is a port-selection problem. Common choices are:

- exhaustive search for small M ,
- greedy add/swap search,
- relaxed binary selection followed by rounding.

Repeat the two steps until the objective improvement is small.

4 Hybrid beamforming in FAS

In hybrid beamforming, the beamformer is split into a digital stage and an analog RF stage:

$$\mathbf{x} = \mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}} \mathbf{s},$$

where

$$\mathbf{W}_{\text{BB}} \in \mathbb{C}^{N_{\text{RF}} \times N_s}, \quad \mathbf{F}_{\text{RF}} \in \mathbb{C}^{N \times N_{\text{RF}}}.$$

The effective beamformer is

$$\mathbf{F} = \mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}}.$$

The power constraint is

$$\text{tr}(\mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}} \mathbf{W}_{\text{BB}}^{\text{H}} \mathbf{F}_{\text{RF}}^{\text{H}}) \leq P.$$

For user k ,

$$y_k = \mathbf{h}_k^{\text{H}}(\mathcal{A}) \mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}} \mathbf{s} + n_k.$$

Hence,

$$\gamma_k = \frac{|\mathbf{h}_k^{\text{H}}(\mathcal{A}) \mathbf{F}_{\text{RF}} \mathbf{w}_k|^2}{\sum_{i \neq k} |\mathbf{h}_i^{\text{H}}(\mathcal{A}) \mathbf{F}_{\text{RF}} \mathbf{w}_i|^2 + \sigma_k^2}.$$

Analog-RF constraints

The analog matrix $\mathbf{F}_{\text{RF}}$ is hardware constrained.

- **Fully-connected:** each RF chain can drive every active port. A common model is

$$|[\mathbf{F}_{\text{RF}}]_{n,r}| = 1, \quad \forall n, r.$$

- **Sub-connected:** each RF chain drives only a subset of active ports. Then disconnected entries are zero, and nonzero entries satisfy the same constant-modulus condition.

Optimization problem

$$\begin{aligned}
& \max_{\mathcal{A}, \mathbf{F}_{\text{RF}}, \mathbf{W}_{\text{BB}}} \sum_{k=1}^K \log_2(1 + \gamma_k) \\
& \text{s.t.} \quad \text{tr}(\mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}} \mathbf{W}_{\text{BB}}^H \mathbf{F}_{\text{RF}}^H) \leq P, \\
& \quad |\mathcal{A}| = N, \quad \mathcal{A} \subseteq \{1, \dots, M\}, \\
& \quad \mathbf{F}_{\text{RF}} \in \mathcal{F}_{\text{RF}}.
\end{aligned} \tag{2}$$

Here $\mathcal{F}_{\text{RF}}$ is the feasible set imposed by the analog hardware.

How to solve it

Again, the standard method is **alternating optimization**, now with three blocks.

- 1) **Fix $\mathcal{A}$ and $\mathbf{F}_{\text{RF}}$, solve $\mathbf{W}_{\text{BB}}$.** Define the effective channel

$$\tilde{\mathbf{H}} = \mathbf{H}(\mathcal{A}) \mathbf{F}_{\text{RF}} \in \mathbb{C}^{K \times N_{\text{RF}}}.$$

Then solve a reduced-dimension digital beamforming problem by ZF, RZF, or WMMSE.

- 2) **Fix $\mathcal{A}$ and $\mathbf{W}_{\text{BB}}$, solve $\mathbf{F}_{\text{RF}}$.** Two standard approaches are:

- direct sum-rate optimization by projected gradient or manifold optimization,
- factorization of a target fully-digital beamformer:

$$\min_{\mathbf{F}_{\text{RF}}, \mathbf{W}_{\text{BB}}} \|\mathbf{F}_\star - \mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}}\|_F^2 \quad \text{s.t.} \quad \mathbf{F}_{\text{RF}} \in \mathcal{F}_{\text{RF}}.$$

With unit-modulus hardware, a standard projection is

$$[\Pi_{\mathcal{F}_{\text{RF}}}(\mathbf{A})]_{n,r} = e^{j\angle[\mathbf{A}]_{n,r}}.$$

- 3) **Fix $\mathbf{F}_{\text{RF}}$ and $\mathbf{W}_{\text{BB}}$, solve $\mathcal{A}$.** This is the same FAS port-selection problem as in digital beamforming.

5 Takeaway

- **Digital beamforming:**

$$\mathbf{x} = \mathbf{W}_{\text{BB}} \mathbf{s}, \quad N_{\text{RF}} = N.$$

- **Hybrid beamforming:**

$$\mathbf{x} = \mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}} \mathbf{s}, \quad N_s \leq N_{\text{RF}} < N.$$

- In both cases, FAS adds a **port-selection layer** on top of a classical beamforming problem.
- The standard solution framework is **alternating optimization**.